

- Q.22 What are the different types of energy source? Write down its advantages and disadvantages of it?
- Q.23 Write a short note on solar energy.
- Q.24 What are the advantages and disadvantages of magneto hydro dynamic convertor?
- Q.25 What are the types on renewable energy sources?
- Q.26 Explain different types of tariff with their advantages.
- Q.27 What are process through which energy can be stored?
- Q.28 Discuss the advantages and disadvantages of geothermal energy.
- Q.29 Write a short note on secondary battery.
- Q.30 Energy storage is an important process. Why?
- Q.31 What are the methodologies for preliminary energy audit?
- Q.32 Explain about the instruments which are helpful in energy audit?
- Q.33 Explain the benefits of energy sources used in agriculture sector.
- Q.34 How energy can be obtained from ocean?
- Q.35 Write a short note on load factor and power factor.

#### SECTION-D

**Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 What is renewable and non-renewable energy source? Write any three examples of both. Explain the concept of MPPT.
- Q.37 Write a short note on:  
a) Geothermal energy b) Tidal energy
- Q.38 How energy efficient technology is used in domestic and industrial sector? Explain it.

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#### 4th Sem / IC, Elect, Power Station Engg., Elect. & Eltx. Engg. Sub : Principles of Energy Management/ Energy sources & Mgmt of Elect. Energy

Time : 3Hrs.

M.M. : 100

#### SECTION-A

**Note:** Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Which of the following is NOT an energy audit instrument?
- Energy meters
  - Power factor meters
  - Wind turbine
  - None of the above
- Q.2 What is the definition of an energy audit?
- A financial analysis of energy expenditures
  - A systematic process to assess and analyze energy consumption
  - A study on renewable energy sources
  - A survey on consumer preferences for energy products
- Q.3 Which of the statements is correct about Solar Energy?
- It is a renewable and conventional source of energy
  - It is a non-renewable and non-conventional source of energy

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- c) It is a renewable and non conventional source of energy  
 d) It is a non renewable source of energy
- Q.4 Which of the following is a characteristic of a secondary cell?  
 a) It cannot be recharged  
 b) It can be recharged and reused  
 c) It is used only once  
 d) Its reaction is irreversible
- Q.5 Geothermal energy is obtained from  
 a) The sun                      b) The Earth's core  
 c) Biomass                      d) Nuclear reactions
- Q.6 What kind of energy does a wind turbine use?  
 a) Kinetic energy              b) Potential energy  
 c) Chemical energy            d) Thermal energy
- Q.7 Energy Resources which are being used for many decades are known as \_\_\_\_  
 a) Conventional energy sources.  
 b) Non conventional energy sources  
 c) Primary energy sources  
 d) Fuel cells
- Q.8 Based on usability, Energy Resources are classified into \_\_\_\_\_.  
 a) Primary secondary and tertiary resources  
 b) Primary and secondary resources  
 c) Primary, second, intermediate and tertiary resources  
 d) Primary, intermediate and secondary resources
- Q.9 Solar cell is generally made up of  
 a) Silicon                      b) Carbon  
 c) Sodium                      d) Magnesium

- Q.10 Which tariff structure is based on the maximum demand of the load?  
 a) Simple Tariff  
 b) Flat Rate Tariff  
 c) Flat Demand Rate Tariff  
 d) Block Rate Tariff

### SECTION-B

**Note:** Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Write any one unit of energy.  
 Q.12 Define tariff.  
 Q.13 Renewable energy is energy derived from natural sources that are replenished at a higher rate than they are consumed. (True / False)  
 Q.14 Write the name of one energy audit instrument.  
 Q.15 Expand MPPT.  
 Q.16 The process of producing energy by utilizing heat trapped inside the earth surface is called as \_\_\_\_ energy.  
 Q.17 Define energy audit.  
 Q.18 \_\_\_\_ is the primary waste product produced by a hydrogen fuel cell.  
 Q.19 The energy obtained from the rise and fall of tides is called tidal energy. (True / False)  
 Q.20 What is fuel cell?

### SECTION-C

**Note:** Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 List any five difference between renewable and non renewable energy.